

CURRENT RESEARCH PROGRESS

Personalized Sign Language to Voice Conversion

Current milestone. The project has progressed through the main stages of localized Sinhala dataset construction, audio preparation, speech-transcription experiments, sign-language feature investigation, and skeletal-feature extraction. Approximately 1,400 Sinhala samples have currently been prepared, with each sample designed to connect the signing video, Sinhala caption, aligned audio, skeletal features, and timing information.

1. Sinhala Speech and Audio Dataset Preparation

Sinhala speech and audio data was prepared from selected source videos containing Sinhala signing content. The active-signing portions of the source videos were identified and separated into individual samples. Sinhala subtitle information was obtained from the corresponding SRT caption tracks, with each SRT entry providing the Sinhala sentence together with its start and end timestamps. A key challenge during this stage was determining reliable boundaries for the active-signing clips. A pause in hand or body movement does not necessarily indicate the end of a sentence. Consequently, some automatically generated clips initially began or ended in the middle of a sentence. To address this, an adaptive motion-based cropping approach was tested rather than relying only on fixed-duration segmentation. The current cropping configuration uses a motion threshold of 0.01, a minimum signing duration of 2 seconds, a minimum idle duration of 1 second, and a smoothing value of 10. The maximum clip duration is 1.8 seconds, with 1-second overlap, no start padding, and 1.2 seconds of end padding. An earlier configuration using fixed 5-second clips with 50% overlap was also tested.

Using this process, approximately 1,400 Sinhala samples have been prepared. Each sample is intended to maintain a consistent relationship between the active-signing video, Sinhala caption, Sinhala audio, skeletal feature sequence, and timing metadata. The dataset is currently undergoing a final alignment review. Samples that still contain incomplete sentence boundaries will be merged, extended, corrected, or removed using the SRT timestamps and Sinhala caption boundaries before the final training dataset is prepared.

2. Audio Preprocessing

The audio processing stage was carried out after the active-signing portions of the source videos had been identified and cropped. Rather than extracting audio from the complete source videos, the audio was extracted from the corresponding cropped signing clips so that each audio segment remained associated with its relevant signing sequence.

The extracted audio segments were then processed using the Whisper speech-recognition model to obtain Sinhala transcriptions. Different Whisper configurations, including the Small and Medium models, were explored during this stage, with a fine-tuned Whisper model considered as a more suitable approach for Sinhala-specific transcription.

In parallel, SRT subtitle files corresponding to the source videos were obtained and examined using their timestamp information. The SRT data was used as an additional reference for relating Sinhala captions to the corresponding video segments and for checking the temporal alignment between the signing content, speech, and text.

A limitation identified during this process was that the SRT caption timing does not always correspond exactly to the timing of the spoken audio. In some cases, a Sinhala caption appears slightly before the corresponding speech begins, or its timing extends beyond the actual spoken segment. Therefore, the SRT timestamps cannot always be treated as an exact representation of the speech boundaries. This timing difference needs to be considered when aligning the cropped signing videos, extracted audio, Whisper transcriptions, and Sinhala captions.

3. Sinhala Speech Transcription Experiments

Whisper-based speech-to-text models were investigated to obtain Sinhala transcriptions from the prepared audio. Different model sizes, including Whisper Small and Whisper Medium, were tested as candidate configurations. The purpose of these experiments was to compare their suitability for the available Sinhala speech data and identify a practical approach for the project. A fine-tuned Whisper model was subsequently considered as the preferred direction for improving Sinhala-specific transcription performance.

Ex:

දරුවෝ විශාල පීඩාවකට පත්වෙලා සිටිනවා මේ පංතිකාමරය තුළ මේ අධ්‍යාපනය නිසා. ඒ වගේම තමයි ගුරුවරු, විදුහල්පතිවරු අධ්‍යාපන පරිපාලන ක්ෂේත්‍රයේ අය විශාල පීඩනයකට පත්වෙලා තියෙනවා

4. How2Sign Dataset Investigation

The How2Sign dataset was investigated as a reference resource for understanding how sign-language video can be represented for machine-learning applications. Particular attention was given to the types of features that can describe a signer and their movement over time.

The feature categories relevant to this project include body pose landmarks, left- and right-hand landmarks, facial landmarks, and ordered frame-level sequences. These representations are important because sign language is expressed through coordinated changes in body posture, hand configuration, hand movement, and facial information rather than through a single static image. The investigation therefore helped establish a reference for the feature-processing pipeline used in the localized Sinhala dataset. It also provided a basis for comparing the current feature representation with alternative representations in future experiments.

5. MediaPipe Holistic Feature Extraction

For the localized Sinhala videos, MediaPipe Holistic was used to extract frame-level landmark information. Instead of passing the complete image pixels directly to the model, the current pipeline represents each frame using a numerical vector containing pose, face, and hand landmarks.

The present configuration contains 1,662 values per frame:

- Pose: 33 landmarks $\times$ 4 values (x, y, z, visibility) = 132 values.
- Face: 468 landmarks $\times$ 3 coordinates (x, y, z) = 1,404 values.
- Left hand: 21 landmarks $\times$ 3 coordinates (x, y, z) = 63 values.
- Right hand: 21 landmarks $\times$ 3 coordinates (x, y, z) = 63 values.
- Total: 1,662 values per frame.

For example, a clip containing 100 frames produces a temporal feature sequence with an approximate shape of $100 \times 1,662$. The ordered sequence allows the model to represent changes in body pose, hand movement, hand shape, and facial information across the duration of a signing clip.

The 1,662-dimensional representation is the current extraction configuration and is not treated as a permanent limitation of the project. Later experiments may investigate reduced facial features, selected joints, derived features such as joint angles, distances or velocities, coordinate normalization, missing-landmark handling, or combinations of skeletal and RGB/video features.

6. NumPy (.npy) Feature Extraction and Validation

The extracted MediaPipe feature sequences were stored as NumPy (.npy) files so that the frame-level sequences could be reused efficiently during model development. The .npy files were inspected to verify the extracted arrays, including their numerical ranges and structural properties. Minimum and maximum feature values were checked as part of this validation process.

This stage is important because the saved feature sequences form one of the principal inputs for the future Sinhala sign-language model. The successful extraction and validation of these files therefore represents a completed and reusable part of the current data-processing pipeline.

7. How2Sign Reference Model Training

Before developing the localized Sinhala model, an existing sign-language research model was trained and evaluated using the How2Sign dataset. Prediction outputs and accuracy values were obtained from the trained model. This experiment served as a reference implementation for the project. It demonstrated that the data-loading, feature-processing, training, prediction, and evaluation stages could be executed as a complete pipeline. The verified prediction and accuracy values will be included in the final experimental-results section when the recorded results are consolidated.

8. Current Project Status

- Dataset: Approximately 1,400 Sinhala samples have been prepared.
- Video segmentation: Active-signing clips have been generated using fixed and adaptive cropping experiments. The current adaptive configuration uses a 0.01 motion threshold, 2-second minimum signing duration, 1-second minimum idle duration, smoothing of 10, 18-second maximum duration, 1-second overlap, no start padding, and 1.2-second end padding.
- Speech transcription: Whisper Small and Whisper Medium have been investigated, with a fine-tuned Whisper approach identified as a promising direction for Sinhala-specific transcription.
- Features: The current localized samples contain 1,662-value MediaPipe Holistic feature vectors for each video frame.
- Feature files: Temporal feature sequences have been extracted, stored as .npy files, and validated.
- Reference training: An existing research model has been trained using How2Sign, providing a working reference for the subsequent Sinhala-model development.

9. Next Stage: Sinhala Model Development

The immediate next stage is to complete the final quality check of the approximately 1,400-sample Sinhala dataset. Each sample must be checked to ensure one-to-one alignment between the signing video, skeletal feature sequence, Sinhala caption, Sinhala audio, and timestamps. Once the dataset has been finalized, it will be divided into training, validation, and test sets. The existing research-model pipeline validated using How2Sign can then be adapted for Sinhala output. Further experiments may compare the current full 1,662-value feature representation with reduced or enhanced feature configurations. The resulting models should be evaluated using appropriate translation and system-level measures, including prediction quality, translation accuracy, BLEU score, and processing latency.

Current Progress Statement

The project has now progressed through the principal stages of localized dataset construction, audio preparation, speech-transcription experimentation, sign-language feature investigation, skeletal feature extraction, and reference-model validation. Approximately 1,400 Sinhala samples have been prepared, with MediaPipe Holistic features, and timestamp-aligned audio. The immediate priority is to complete the final alignment review, finalize the verified dataset, and proceed with training a dedicated model for the SLSL-to-Sinhala translation task.